Parametric Equations: Cram Sheet

Posted: 4 questions, 30 minutes total, 12 points, 3 without a calculator and 1 with. The four types below are a **practice framework** built from the notes and worksheets; the real question types and order were not posted.

How to write the final rectangular answer (one rule set, from the homework answer keys)

the curve	write it as	example
one y for each x	solve for y	$y = 2x^2 - 2,\ y = \frac{8}{x}$
one hyperbola branch	$y = \sqrt{\cdot}$ (bottom: $-\sqrt{\cdot}$)	$y = \sqrt{\frac{x^2}{4} + 4}$
whole circle or ellipse	leave it "$= 1$"	$\frac{(x-1)^2}{4} + \frac{(y+2)^2}{9} = 1,\ x^2 + y^2 = 16$
sideways parabola	solve for x	$x = \frac{(y-3)^2}{4}$
expanded or factored	either; don't convert	$y = x^2 - 2x - 3$ or $y = (x-1)^2 - 4$

Always add the restriction (like $0 \le x \le 2$), even though some keys leave it off. The quiz format itself was not posted.

The four types

1 Eliminate t, restrict, graph

1. Solve the simpler equation for t.
2. Substitute; simplify.
3. t-table: ends + any turning point.
4. State the x and y values **actually reached**.
5. Graph with dots and direction arrows.

2 Trig curve

1. Isolate: $\cos t = \frac{x-h}{a}$, $\sin t = \frac{y-k}{b}$ (or $\tan, \sec$ / $\cot, \csc$).
2. Square and use the identity.
3. Pick the branch: $\sec t > 0$ on $(-\frac{\pi}{2}, \frac{\pi}{2})$, $\csc t > 0$ on $(0, \pi)$.
4. t-table for direction.

3 Two parametrizations

1. $x = t$, $y = f(t)$.
2. Complete the square: $y = (x-h)^2 + k$.
3. $t = x - h$, so $x = t + h$, $y = t^2 + k$.
4. Write "for t in $(-\infty, \infty)$" with both (when every x is allowed).

4 Projectile (calculator, degree mode)

$x = (v\cos\theta)t, \quad y = -16t^2 + (v\sin\theta)t + h$

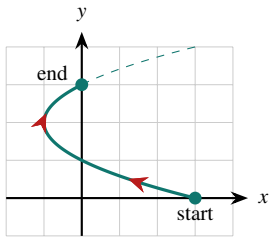
1. Flight time: $y = 0$, quadratic formula, reject negative t.
2. Distance: **unrounded** t into x.
3. Top: $t = \frac{v\sin\theta}{32}$, then y.
4. Rectangular: $t = \frac{x}{v\cos\theta}$ ($v\cos\theta \neq 0$).

Identities

you see	use	you get
$\sin t$, $\cos t$	$\sin^2 t + \cos^2 t = 1$	circle or ellipse
$\tan t$, $\sec t$	$\sec^2 t - \tan^2 t = 1$	hyperbola, upper branch on $(-\frac{\pi}{2}, \frac{\pi}{2})$
$\cot t$, $\csc t$	$\csc^2 t - \cot^2 t = 1$	hyperbola, upper branch on $(0, \pi)$
$\sin t$, $\csc t$	$\csc t = \frac{1}{\sin t}$	$y = \frac{k}{x}$ piece

Arrows, ends and restrictions

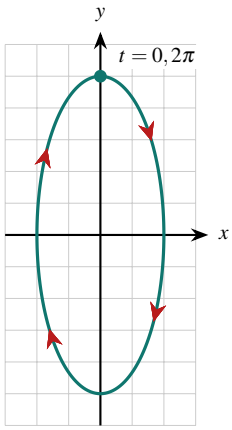
- **Red arrows along the curve**: the way t increases. Check with a t-table: start, middle, end.
- **Arrowhead at the very end**: the curve keeps going (x or $y \to \pm\infty$).
- **Solid dot**: endpoint reached (closed bracket).
- **Open circle**: excluded endpoint, *only if* no other allowed t reaches that point.
- **Closed curve**: an ellipse over $[0, 2\pi]$ starts and ends at one point.
- $\sqrt{\ }$ gives $x \geq 0$. t^2, $\sin t$, $\cos t$ can turn around inside the interval: check the middle. Carry $t \neq c$ over to x.



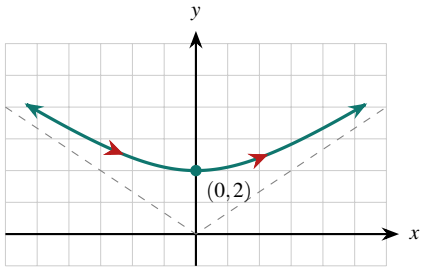
$x = t^2 - 1,\ y = t + 2,\ [-2, 1]$
$-1 \leq x \leq 3,\ 0 \leq y \leq 3$

Curve shapes and direction

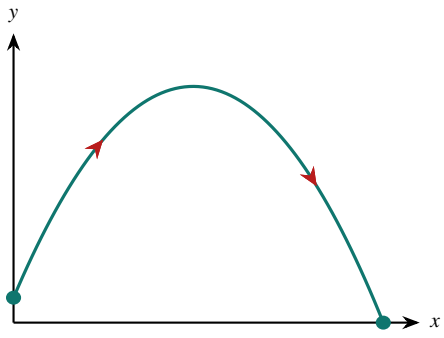
sideways parabola	$x = (y-k)^2 + h$, from $x = t^2$ and y linear in t
parabola up/down	$y = a(x-h)^2 + k$, from x linear in t and $y = t^2$; projectiles open down
circle, ellipse	$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$. With $a, b > 0$: $a\cos t,\ b\sin t$ counterclockwise from $(a, 0)$; $a\sin t,\ b\cos t$ clockwise from $(0, b)$. One sign flip reverses it; two keep it.
hyperbola branch	$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$; final answer $y = \sqrt{\cdot}$. tan / sec: left to right. cot / csc: right to left. Asymptotes $y = \pm\frac{a}{b}x$.
reciprocal	$y = \frac{k}{x}$; $x = 2\sin t$, $y = 4\csc t$ on $(0, \pi)$ retraces: down to $(2, 4)$, back up.



$x = 2\sin t,\ y = 5\cos t$
clockwise, one start/end



$x = 3\tan t,\ y = 2\sec t$
upper branch $y \geq 2$, left to right



$v = 128,\ 24°,\ h = 5$: lands $t \approx 3.347$,
$x \approx 391.407$ ft

Red arrows: direction as t increases. Plain arrowheads at the ends of the hyperbola: the curve keeps going.

Traps

- Only plugging the t endpoints into x when x turns around inside the interval.
- Both hyperbola branches, or $y = \pm\sqrt{\cdot}$, when $\sec t$ or $\csc t$ is always positive.
- Leaving off the t interval on a parametrization; $t = x + 3$ means $x = t - 3$.
- Rounding t before finding x; radian mode with degree angles.

Sources: combined parametric notes, day 1 and day 2 worksheets, warm-up after day 1, the Before Parametrics page (conics). Separate day 2 notes and the warm-up before parametrics (PA) were not available.